

In 1834, Heinrich Lenz, a Russian physicist, formulated a law to describe the direction of induced electric current. He used logic and the law of conservation of energy to deduce the direction of an induced current. You can observe the direction of current by watching a needle on a galvanometer move as you move a magnet along a conductor. Lenz determined how to predict that direction.

Mini Investigation

Observing the Direction of an Induced Current

Skills: Predicting, Performing, Observing, Analyzing

SKILLS
HANDBOOK  A2.1, A2.4

In this activity, you will observe the direction of the induced current when moving a magnet into and out of a coiled conductor.

Equipment and Materials: galvanometer; coiled conductor; permanent bar magnet; 2 alligator clip leads

1. Connect the two terminals of the coil to the galvanometer.
2. At a moderate rate, push the north pole of the magnet into the coil (**Figure 1**). Note the direction of the current on the galvanometer.

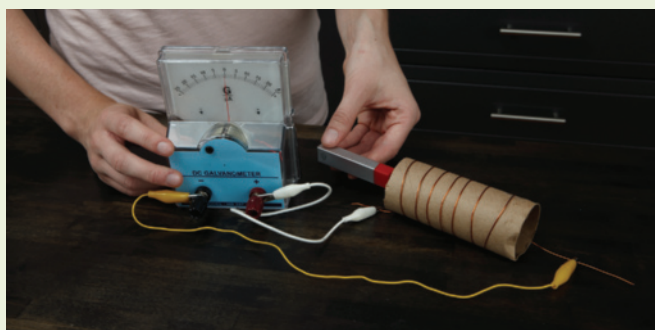


Figure 1

3. Pull the north pole of the magnet out of the coil and note the direction of the current on the galvanometer.
4. Repeat Steps 2 and 3 with the south pole of the magnet.
 - A. How did the direction of the current compare in Steps 2 and 3? [T/I](#)
 - B. Describe the direction of the current in Step 4 compared to Steps 2 and 3. [T/I](#)
 - C. Did using the south pole instead of the north pole change your results? Explain. [T/I](#)
 - D. Predict the magnetic pole induced at the end of the coil nearest the magnet (using the right-hand rule for a coiled conductor) for both Steps 2 and 3. [T/I](#)

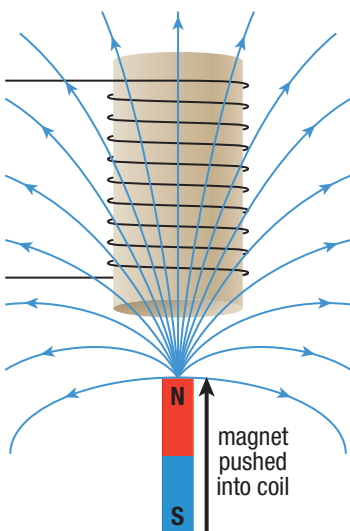


Figure 2 North pole of a permanent magnet pushed into a coiled conductor

Direction of Induced Current

When a permanent magnet is pushed into a coiled conductor, the electrons in the coil respond to the magnetic field by starting to move in a particular direction and forming an electric current. The direction of the current depends on the direction in which the magnetic field points. For example, if the north pole of a magnet is pushed into a coil, an induced current is produced in the coil in one direction (**Figure 2**). If the south pole of the magnet is pushed into a coil, the induced current is in the opposite direction.

How did Lenz deduce the direction of an induced current? He used the law of conservation of energy as a starting point. Recall from Chapter 5 that energy cannot be created or destroyed. All that can be done is to transform one type of energy into another. When the north pole of a magnet is moved into the coil, as in Figure 2, kinetic energy exists in the movement of the magnet. The kinetic energy is transformed into electrical energy in the electric current in the coil. As a result, the electric current in the coil also produces a magnetic field. Which way does the magnetic field point? Is the end of the coil nearest the bar magnet in Figure 2 a north magnetic pole or a south magnetic pole?

Let us first consider that the conventional induced current goes around the coil from left to right at the front of the coil in Figure 2. In this case, the right-hand rule for a solenoid would indicate that the bottom of the coil is a south magnetic pole. However, this is inconsistent with the law of conservation of energy! If the bottom of the coil were a south magnetic pole, then it would attract the north pole of the magnet into the coil without the need for an external force to push the magnet into the coil. So the bottom of the coil cannot be a south magnetic pole.

Now consider that the conventional induced current goes around the coil from right to left at the front of the coil. The right-hand rule for a solenoid would indicate that the bottom of the coil is a north magnetic pole. The permanent magnet is then repelled by the north magnetic pole of the coil. Therefore, you need to apply a force to push the permanent magnet into the coil, thereby doing work. You are transforming kinetic energy into electrical energy in the coil. This idea is consistent with the law of conservation of energy.

With similar deductive reasoning, Lenz was able to summarize his findings into a law for determining the direction of an induced current, known as **Lenz's law** (Figure 3).

Lenz's law if a changing magnetic field induces a current in a coil, the electric current is in such a direction that its own magnetic field opposes the change that produced it

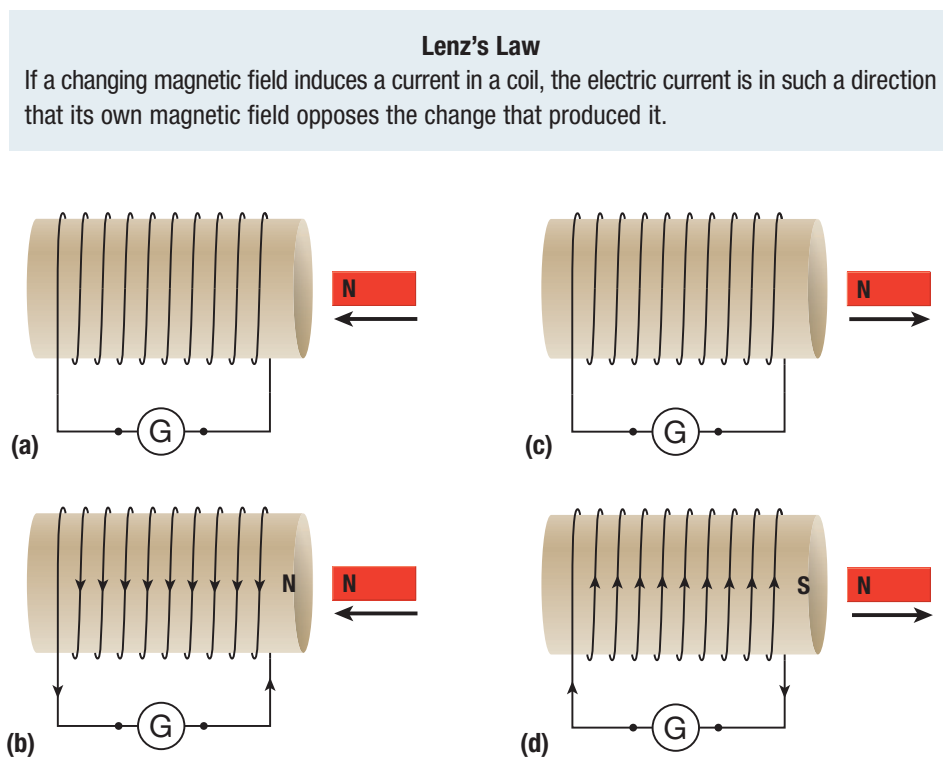


Figure 3 Lenz's law

- (a) The permanent magnet is being forced into the coil. If north is being forced inward, the coil must oppose that with a north pole on the right side of the coil.
- (b) If the right side of the coil must be north, then the right-hand rule for a solenoid determines the direction of the electric current to be down the coil at the front.
- (c) The permanent magnet is being pulled out of the coil. If north is being forced to the right, the coil must oppose that with a south pole at the right side of the coil.
- (d) If the right side of the coil must be south, then the right-hand rule for a solenoid determines the direction of the electric current to be up the coil in the front.

LEARNING TIP

Remembering Lenz's Law

The coil opposes whatever the magnet is trying to do. If north is moving in, the coil repels it with a north. If north is moving out, the coil attracts it back with a south.

Lenz's Law and Drop-Tower Rides

Drop-tower rides are popular attractions at amusement parks (**Figure 4**). Riders are strapped into a seat and can be lifted to a height of over 70 m. They are then released to free fall toward the ground. To prevent disaster, the braking system must be extremely reliable. If a friction-based braking system were used, it would need to be triggered at just the right time, and the brakes would wear out quickly and need constant replacement. Drop-tower rides use an ingenious system that relies on electromagnetic induction. It can be explained using Lenz's law.



Figure 4 Drop-tower rides use an electromagnetic braking system that is reliable and automatic and has parts that do not wear out.

Each cart on the ride has permanent magnets under the seats. After approximately 45 m of free fall, an electromagnetic braking system kicks in. Along the bottom third of the tower are copper strips mounted vertically on the tower. When the cart falls and the permanent magnets move past the copper conductor, an electric current is induced in the copper. The induced current then produces a magnetic field. Applying Lenz's law, the induced magnetic field must oppose the field that created it. The opposing repulsion force acts to create a reliable, no-friction braking system.

Investigation 13.2.1

Investigating Electromagnetic Induction (p. 613)

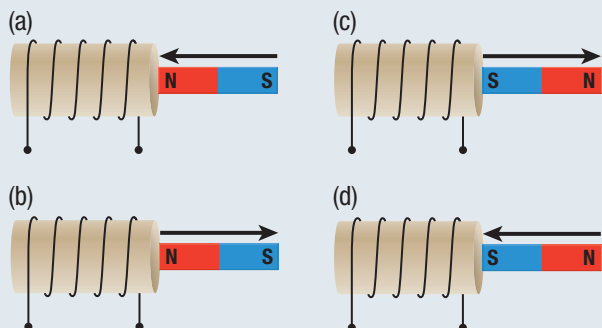
Now that you have learned about electromagnetic induction, you can do some further observation by completing Investigation 13.2.1.

13.2 Summary

- Lenz's law states that if a changing magnetic field induces a current in a coil, the electric current is in such a direction that its own magnetic field opposes the change that produced it.

13.2 Questions

1. For each diagram below, determine the direction of the induced current in the coil and the magnetic poles on the coil. **T/I**



2. How would your answers to Question 1 change if the coils were moved and the magnets were stationary? Assume the coils were moved in the opposite direction to the arrows shown. **K/U**
3. When a magnet is pushed into a coil to induce a current, the magnetic field that is created never attracts the magnet into the coil. Explain why this is the case. **K/U**
4. In a drop-tower ride, would using an electromagnet in the carts, instead of the permanent magnets under the seats, work equally well? Would it be equally reliable? Explain. **A**